

Arrays

1. **Prefix Sum (Cumulative Sum)**
2. **Kadane's Algorithm** (Maximum Subarray Sum)
3. **Binary Search**
4. **Two-Pointer Technique**
5. **Sliding Window**

Linked Lists

1. **Reversal of a Linked List**
2. **Detecting a Cycle** (Floyd's Tortoise and Hare)
3. **Merging Two Sorted Linked Lists**
4. **Finding the Middle of a Linked List**
5. **Remove N-th Node from End**

Stacks

1. **Balanced Parentheses Checking**
2. **Infix to Postfix Conversion**
3. **Next Greater Element**
4. **Stock Span Problem**
5. **Largest Rectangle in Histogram**

Queues

1. **Breadth-First Search (BFS)**
2. **Sliding Window Maximum**
3. **Implementing a Queue using Stacks**

Trees

1. **Depth-First Search (DFS)**
2. **Breadth-First Search (BFS) / Level Order Traversal**
3. **Inorder, Preorder, Postorder Traversal**
4. **Binary Search Tree (BST) Insertion, Deletion, Search**
5. **Lowest Common Ancestor (LCA)**
6. **AVL Tree Rotations**
7. **Segment Trees (Range Query, Point Update)**
8. **Fenwick Tree (Binary Indexed Tree)**

Heaps

1. **Heapify**
2. **Heapsort**
3. **Priority Queue Operations**
4. **Median Maintenance using Heaps**

Hashing

1. **Hash Table Operations (Insert, Delete, Search)**
2. **Counting Frequencies of Elements**
3. **Detecting Duplicates in Linear Time**
4. **Finding Pair with Given Sum**

Graphs

1. **Depth-First Search (DFS)**
2. **Breadth-First Search (BFS)**
3. **Dijkstra's Algorithm (Shortest Path)**
4. **Floyd-Warshall Algorithm (All-Pairs Shortest Path)**
5. **Prim's and Kruskal's Algorithms (Minimum Spanning Tree)**
6. **Bellman-Ford Algorithm**
7. **Topological Sorting**
8. **Strongly Connected Components (Kosaraju's Algorithm)**
9. *A Search Algorithm**

Strings

1. **Knuth-Morris-Pratt (KMP) Algorithm**
2. **Rabin-Karp Algorithm**
3. **Z Algorithm**
4. **Longest Common Subsequence (LCS)**
5. **Longest Palindromic Substring**
6. **Trie Operations (Insert, Search, Delete)**
7. **Suffix Array and Suffix Tree Construction**

Dynamic Programming (often applied across multiple data structures)

1. **Longest Increasing Subsequence (LIS)**
2. **Edit Distance**
3. **0/1 Knapsack Problem**
4. **Coin Change Problem**
5. **Matrix Chain Multiplication**
6. **Fibonacci Sequence**

Miscellaneous

1. **Union-Find / Disjoint Set Union (DSU)**
2. **Bit Manipulation Techniques**
3. **Reservoir Sampling**